

Chitragupta

Operations Handbook — costs, memory, limitations, roadmap

Running costs

The only per-use cost is the Anthropic API (Claude Sonnet). Planning estimates at typical Sonnet pricing (~\$3/M input, ~\$15/M output tokens) — verify current rates at anthropic.com/pricing:

Interaction	Typical tokens	Est. cost each
How-to / explain	1–2k in, 300 out	< \$0.01
Context Q&A on a document	3–5k in, 300 out	\$0.01–0.02
Analytics over ≤200 rows	5–15k in, 500 out	\$0.02–0.06
Document creation (multi-turn)	2–4 model calls	\$0.02–0.05
Agent request (≤8 steps)	8–20k total	\$0.05–0.25

Pilot budget: 20 users × ~15 messages/day ≈ **\$5–15/day** (■400–1,250), roughly **\$150–450/month**. Levers: CHITRAGUPTA_MODEL=haiku for planning (~5–10× cheaper), cache how-to answers, lower the analytics row cap. Infrastructure is self-hosted — no other per-seat SaaS fees.

Memory model — what persists vs what clears

Layer	Stored where	Backend restart	Page refresh
Personal preferences (“remember X”)	Database	kept	kept
Org norms learned from your edits	Database	kept	kept
Skills / audit / pending proposals	Database	kept	kept
Conversation context (12 turns/user)	Backend RAM	cleared	kept
Chat display in the widget	Browser sessionStorage	kept	kept (same tab); cleared on new tab / browse
Act-as-user session bindings	Backend RAM	cleared — users just reload the ERP page	kept
Backend login tokens	Database (12h TTL)	kept until expiry	kept

Rule of thumb: **learned knowledge is permanent; conversational context is ephemeral**. A restart never loses learning, audit, or pending approvals.

Honest limitations

- Analytics row cap: ≤200 rows fed to the model (≈60 if very wide). Company-scale tables need server-side aggregation (roadmap #2).
- Attachments: first 3 files, ~6k chars each; NO OCR — scanned/image PDFs and photos aren't read yet.
- No streaming: complex answers take 3–10s (thinking indicator shown).
- English only today; Hindi/Gujarati on the roadmap.
- Approvals use Chitragupta's own gate; native ERPNext Workflow chains not yet driven (roadmap #1).
- The model can be wrong: final wording/arithmetic is model-generated from real rows — the design assumes human review, and no write ever bypasses approval.
- Single process is right for a 20-user pilot; scale-out later moves session bindings to DB/redis.
- “Bug-free” is not an honest claim. What IS true: 131 tests pass, every live-found bug has a regression test, and the system fails safe.

Learning — how it grows, how to verify it

- Edit a draft before approving → that correction becomes an org norm → the NEXT draft applies it automatically (“applied learned default ...” in the activity log).
- Clean approvals reinforce supplier/item rates as precedents for future drafts.
- “remember default_warehouse Unit-2” → personal preference, visible & deletable.
- Every task type tracks runs vs approved-without-edits.
- **The metric:** chart /api/learning clean_rate weekly. Rising = genuinely learning your company. Model weights never change — all learning is inspectable, attributable, deletable (by design: silent drift would be a governance nightmare).

Troubleshooting quick table

Symptom	First check	Usual fix
Widget missing	browser console; /assets/chitragupta_desk/... 404	install-app + bench build (bench RUNNING) + clear-cache + hard re
Canned “I didn’t follow that”	boot log says “rules only”	ANTHROPIC_API_KEY missing in that shell
Wrong identity / “Purchase User”	backend log: /api/login not /api/sso	SSO chain: api.py, matching secret both sides, clocks, fresh widget
SSO 401	the 401’s detail now states the reason	clock sync or secret mismatch as stated
403 from ERP	which user does the error name?	if bot: role fence working (correct); if real user: their ERPNext perm
Same bug “again”	/api/version build stamp	stale deploy — redeploy the piece that’s old

Pending roadmap (priority order)

#	Item	Why
1	ERPNext Workflow chains	Company’s real multi-level approvals drive routing
2	Server-side aggregation	Remove the row cap for company-scale analytics
3	Streaming (SSE)	Perceived speed
4	WhatsApp channel	Approvals-on-phone first; where voice matters
5	Multilingual (Hindi/Gujarati)	Shop floor
6	Automode + vetted skills	High-clean-rate tasks graduate to supervised autonomy, kill switch
7	OCR/vision	Scanned attachments
8	DB-backed sessions + scale-out	Beyond one process

Recommendations

- Rotate EVERY credential used during development (they appeared in chats/screenshots) before real data.
- Pilot 20 Purchase users drafts-only; expand on demonstrated adoption, not features.
- Chart clean_rate weekly — it is your “is it learning?” truth and your director metric.
- Never widen the bot’s role to silence an error; each 403 so far was the safety model working.
- Check /api/version first whenever something “old” reappears — deploy drift caused more ghost-bugs than code did.
- Fix clock sync properly (NTP, disable hypervisor sync) — audit timestamps depend on it.